

Accurate Doppler Frequency Shift Estimation for any Satellite Orbit

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Abstract- One of the most important parameters in satellite communication links, particularly in Low Earth Orbits (LEO) and elliptical orbits is Doppler Frequency Shift (DFS) and Doppler Frequency Rate (DFR) at communication channels. This paper analyzes Doppler frequency effects in various satellite orbits. For this purpose, a new orbit generator is presented for accurate simulation of satellite orbits. The non-spherical mass distribution turbulence of the earth and its effects on Doppler shift calculations is considered and simulation results of Doppler frequency shift for an elliptical LEO satellite in L-band is presented.

I. INTRODUCTION

In communication systems in which receiver and transmitter are not fixed and have significant relative velocity, Doppler frequency shift is high. For proper design of various parts of transceivers like input filters in receivers and PLL parts, we need to accurately calculate carrier frequency shift and compensate it.

With the designed simulator, receivers can calculate time interval of viewing satellite and Doppler frequency in each period. Accurate Doppler information improves the phase lock loop performance and reduces power consumption of ground segment.

There are many researches which have focused on methodology to compensate for Doppler shifts but only a few exist on accurate estimation of this parameter. Doppler time curves in the simple case of a circular orbit in the equatorial plane and Doppler observation points on the equator have been analyzed in Ref.[1]. In Ref.[2] the author derived analytical expression of Doppler shift for circular LEO satellites. In this paper we introduce a scheme for any satellite orbit and this simulator is designed in MATLAB simulink environment.

II. ORBIT DYNAMICS

Among many kinds of nongeostationary orbits, there are two typical orbits for the communication satellites; highly elliptical and low-altitude orbits. Since the nongeostationary satellites change their relative positions to earth stations, communication signals suffer from

Doppler shift. The value and drift rate of Doppler shift are very large compared with that in geostationary satellites or terrestrial systems. One consequence of the use of elliptical orbits is the existence of Doppler shifts in all satellite/ ground link.

To calculate the accurate Doppler frequency shift, we need the relative velocity between satellite and ground terminal. The relationship between the relative velocity and Doppler shift is given by:

$$f_d = \frac{f_0 v_t}{c} \text{ Where } f_0 \text{ is the carrier frequency and}$$
$$v_t = \frac{d(P_s - P_e)}{dt} \text{ in spherical coordinates}$$

Where P_s and P_e are the position of satellite and earth transceiver respectively.

To calculate this relative velocity, an orbit generator is presented here. This estimator uses Kepler's equation to determine position and velocity of satellite in orbit. To determine an earth orbit we need six parameters, the parameters are defined at some reference time or epoch (t_0): *semimajor axis* (a) and *eccentricity* (e) that define the size and shape of the orbit, the *inclination* (i) and *right ascension of the ascending node* (Ω) that define the orbit plane, the rotation of the orbit within the plane which is defined by the *argument of perigee* (ω), and finally the *mean anomaly* (M) specifies the position of the satellite in its orbit at the epoch time.

In figure 1 if the plane of paper is the reference plane and the dashed part of orbit is below the paper, then the nodes are as illustrated.

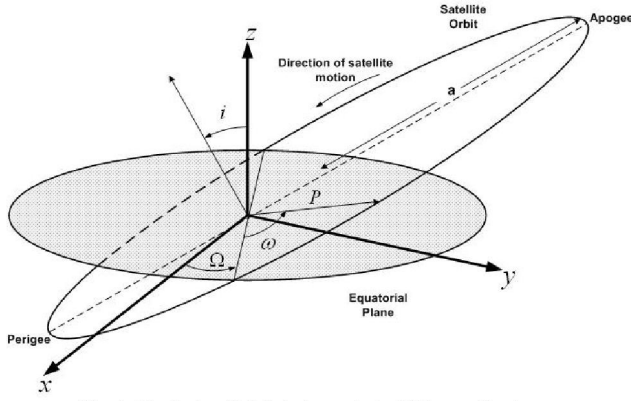


Fig. 1. Keplerian Orbital elements in ECI coordinate

To determine the relative velocity we need to calculate velocity of satellite and ground station in a set of reference coordinates. For this purpose the satellite velocity is calculated in ECI coordinates and is then converted to ECF coordinates; ground station velocity is also determined in ECF coordinates.

Different perturbing forces like Nonhomogeneity and oblateness of the earth affect the orbit parameters as follow:

Rate change of orbit parameters influenced by perturbing forces are represented by Gauss planetary equations along the axes of a moving Cartesian frame defined in the following way: R along the radius vector r ; S in the local plane of osculating orbit, perpendicular to R , and in direction of satellite motion; W perpendicular to both R and S , in the direction of the momentum vector $R \times S$. any perturbing force can then be expressed as:

$$\mathbf{r}_p = R\mathbf{R} + S\mathbf{S} + W\mathbf{W} \quad (4)$$

$$K = -1.5 * \mu * J_2 * R_E^2 / r^4 \quad (5)$$

$$R = K \left(1 - 3 \sin^2(\omega + \nu) \sin^2(i) \right) \quad (6)$$

$$S = K \sin(2(\omega + \nu) \sin^2(i)) \quad (7)$$

$$W = K \sin(\omega + \nu) \sin(2i) \quad (8)$$

The resulting Gauss equations are:

$$\frac{da}{dt} = \frac{2}{na\sqrt{1-e^2}} (eR \sin \nu + (1+e \cos \nu)S) \quad (9)$$

$$\frac{de}{dt} = \frac{\sqrt{1-e^2}}{na} (R \sin \nu + (\cos E + \cos \nu)S) \quad (10)$$

$$\frac{di}{dt} = \frac{1}{na\sqrt{1-e^2}} \frac{r}{a} \cos(\nu + \omega)W \quad (11)$$

$$\frac{d\Omega}{dt} = \frac{1}{na\sqrt{1-e^2}} \frac{r}{a} \frac{\sin(\nu + \omega)}{\sin i} W \quad (12)$$

$$\frac{d\omega}{dt} = \frac{\sqrt{1-e^2}}{nae} \left(-R \cos \nu + \left(1 + \frac{1}{1+e \cos \nu} \right) S \sin \nu - \frac{d\Omega}{dt} \cos i \right) \quad (13)$$

$$\frac{dM}{dt} = n + \frac{1-e^2}{nae} \left(R \left(\frac{-2e}{1+e \cos \nu} + \cos \nu \right) - S \left(1 + \frac{1}{1+e \cos \nu} \right) \sin \nu \right) \quad (14)$$

Equations 9-14 show if the perturbing force vector is known, then the differential changes of all six orbit parameters can be calculated analytically.

III. ORBIT GENERATOR

satellite position and velocity can be derived from orbit parameters.

The mean motion is computed from semi major axis:

$$n = \sqrt{\frac{\mu}{a^3}}, \quad \mu = 3.986005 \text{e}14 \quad (15)$$

Mean anomaly in term of mean motion is:

$$M = n(t - T) \quad (16)$$

Eccentric anomaly can be computed from mean anomaly and eccentricity.

$$E = M + e \sin M + \frac{1}{2} e^2 \sin 2M + \frac{1}{8} e^3 (3 \sin 3M - \sin M) + \dots \quad (17)$$

With knowing the eccentric anomaly, true anomaly is calculated from the equation below:

$$\tan \frac{\nu}{2} = \left[\frac{(1+e)}{(1-e)} \right]^{1/2} \tan \frac{E}{2} \quad (18)$$

distance of satellite from the center of earth is:

$$r = \frac{a(1-e^2)}{(1+e \cos(\nu))} \quad (19)$$

With the calculated parameters, satellite position in ECI coordinates is represented as bellow:

$$\begin{bmatrix} X \\ Y \\ Z \end{bmatrix} = r \begin{bmatrix} \cos(\omega + \nu) \cos(\Omega) - \sin(\omega + \nu) \sin(\Omega) \cos(i) \\ \cos(\omega + \nu) \sin(\Omega) + \sin(\omega + \nu) \cos(\Omega) \cos(i) \\ \sin(\omega + \nu) \sin(i) \end{bmatrix} \quad (20)$$

Satellite ECI velocity drives from differentiation of position with respect to time.

$$\begin{bmatrix} V_x \\ V_y \\ V_z \end{bmatrix} = \frac{na}{r} \begin{bmatrix} bl_2 \cos E - al_1 \sin E \\ bm_2 \cos E - am_1 \sin E \\ bn_2 \cos E - an_1 \sin E \end{bmatrix} \quad (21)$$

Where:

$$b = a(1 - e^2)^{1/2} \quad (22)$$

$$l_1 = \cos \Omega \cos \omega - \sin \Omega \sin \omega \cos i \quad (23)$$

$$m_1 = \sin \Omega \cos \omega + \cos \Omega \sin \omega \cos i \quad (24)$$

$$n_1 = \sin \omega \sin i \quad (25)$$

$$l_2 = -\cos \Omega \sin \omega - \sin \Omega \cos \omega \cos i \quad (26)$$

$$m_2 = -\sin \Omega \sin \omega + \cos \Omega \cos \omega \cos i \quad (27)$$

$$n_2 = \cos \omega \sin i \quad (28)$$

IV. DOPPLER EQUATIONS

To calculate the relative velocity we need to have velocity vector of satellite and ground station in the same coordinate system. position of ground station in ECF coordinates can be derived easily by knowing the longitude and latitude of station and hence by transforming the satellite coordinate from ECI to ECF we reach the goal.

The transformation of an ECI position vector r_{ECI} to an ECF position vector r_{ECF} is given by the following vector-matrix operation $r_{ECF} = [T]r_{ECI}$ where the elements of the transformation matrix $[T]$ are given by

$$[T] = \begin{bmatrix} \cos \theta & \sin \theta & 0 \\ -\sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (29)$$

where θ is the Greenwich sidereal time at the moment of interest. Greenwich sidereal time is given by the following expression:

$$\theta = \theta_{g0} + \omega_e t \quad (30)$$

where θ_{g0} is the Greenwich sidereal time at 0 hours UT, ω_e is the inertial rotation rate of the Earth, and t is the elapsed time since 0 hours UT.

The ECF velocity vector is determined by differentiating this expression:

$$V_{ECF} = [T]\dot{r}_{ECI} + [\dot{T}]r_{ECI} = [T]V_{ECI} + [\dot{T}]r_{ECI} \quad (31)$$

The elements of the $[\dot{T}]$ matrix are as follow:

$$[\dot{T}] = \begin{bmatrix} -\omega_e \sin \theta & \omega_e \cos \theta & 0 \\ -\omega_e \cos \theta & -\omega_e \sin \theta & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad (32)$$

If latitude and longitude of ground segment are φ, λ then the ground segment position in ECF will be:

$$\begin{bmatrix} x_G \\ y_G \\ z_G \end{bmatrix} = R_E \begin{bmatrix} \cos \lambda \cos \varphi \\ \sin \lambda \cos \varphi \\ \sin \varphi \end{bmatrix} \quad (33)$$

With knowing the position vector of satellite and earth terminal in ECF coordinates, the position vector of satellite with terminal reference can be derived. With this vector and the satellite velocity vector, relative velocity of satellite with respect to earth terminal in line of sight view can be derived.

In figure 2, $\vec{r}_1$ is the earth terminal position vector and $\vec{r}_2$ is the satellite position vector. Hence $\vec{r}_3$ is defined as the relative position of satellite with respect to earth terminal in ECF coordinates. Relative velocity between satellite and ground terminal is: $\vec{V}_{rel} = \vec{V}_{sat} - \hat{\vec{r}}_3$ where $\hat{\vec{r}}_3 = \frac{\vec{r}_3}{|\vec{r}_3|}$ and $\vec{r}_3 = \vec{r}_1 - \vec{r}_2$

$$\vec{r}_1 = \begin{bmatrix} R_E \cos \lambda \cos \varphi \\ R_E \sin \lambda \cos \varphi \\ R_E \sin \varphi \end{bmatrix}^T \begin{bmatrix} \hat{x} \\ \hat{y} \\ \hat{z} \end{bmatrix} \quad (34)$$

$$\vec{r}_2 = r \begin{bmatrix} \cos(\omega + \nu) \cos(\Omega + \theta) - \sin(\omega + \nu) \sin(\Omega + \theta) \cos i \\ \cos(\omega + \nu) \sin(\Omega + \theta) + \sin(\omega + \nu) \cos(\Omega + \theta) \cos i \\ \sin(\omega + \nu) \sin i \end{bmatrix}^T \begin{bmatrix} \hat{x} \\ \hat{y} \\ \hat{z} \end{bmatrix} \quad (35)$$

In the above equations $\vec{V}_{sat}$ is satellite velocity in ECF coordinates. By knowing the accurate relative velocity, Doppler frequency shift can be calculated.

Another problem seen in the above equations is that Doppler shift is calculated for the whole time but we want to compute this parameter only in the visible time intervals. For this purpose, by using the minimum earth elevation angle, the satellite visibility intervals can be determined.

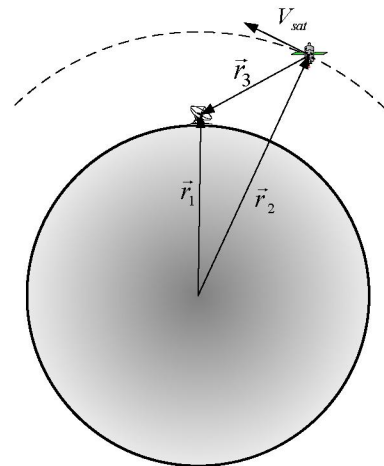


Fig. 2. Position and velocity vectors of satellite and earth terminal

V. SIMULATION RESULTS

Analytical results are simulated in MATLAB simulink environment.

Ground terminal has 51.447651 degree north longitude and 35.774475 degree east latitude. Earth elevation angle is 0.

Orbit parameters are as below:

- mean anomaly = 0
- semi major axis = 7678137.085 m
- inclination = 105°
- eccentricity = 0.1
- argument of perigee = 270°
- RAAN = 155°

Figure 3 shows the ECI satellite position for 2 hours.

Figure 4 shows ECF satellite velocity.

Doppler frequency in satellite visibility intervals is shown in figure 5. It shows that in a course of a day the satellite can communicate with the ground terminal in 9 visibility intervals.

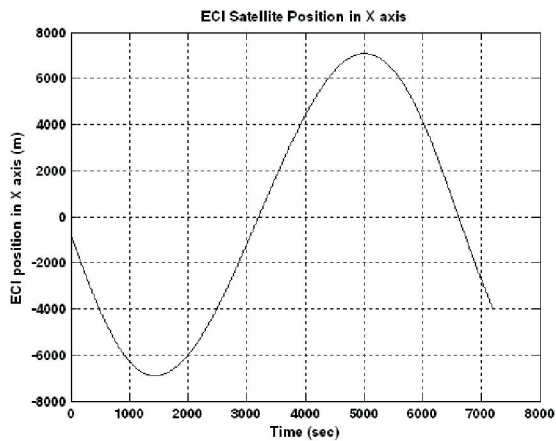


Fig. 3. ECI satellite position for 2 hours

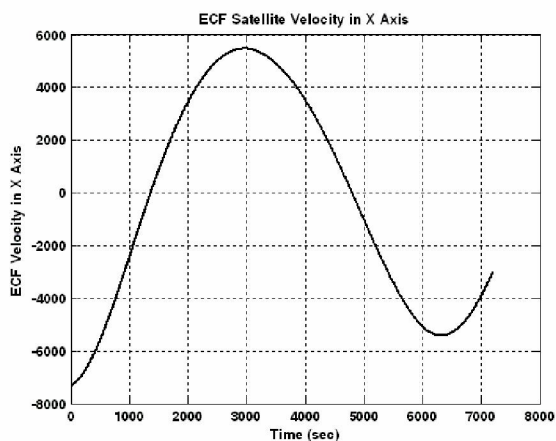


Fig. 4. ECF satellite velocity

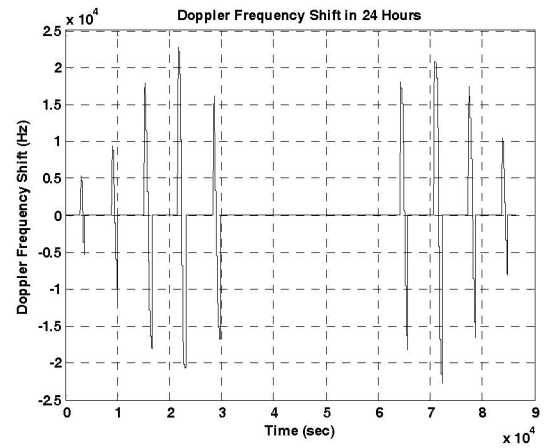


Fig. 5. Doppler frequency shift in 24 hours

In the fourth pass of satellite (zoomed in figure 6), positive and negative Doppler shifts are not the same due to elliptical orbit.

With changing the earth elevation angle, satellite visibility intervals also change, in the figures 7 and 8, Doppler shift for 5° and 20° of earth elevation angle are presented. As seen, by increasing the elevation angle, visibility intervals decrease.

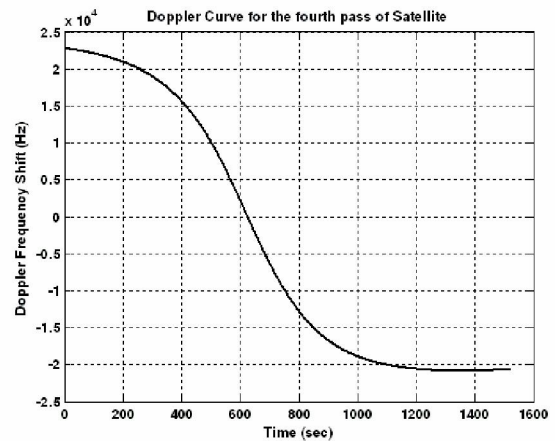


Fig. 6. MATLAB results for the forth pass

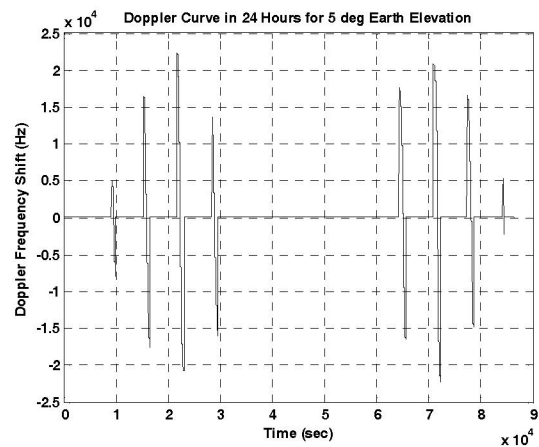


Fig. 7. Doppler shift for 5 degree earth elevation in a day

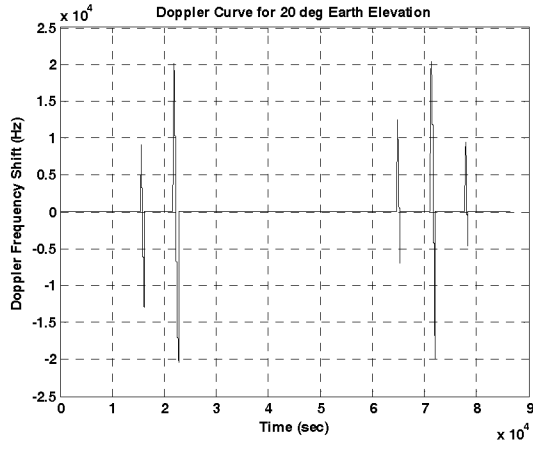


Fig. 8. Doppler shift for 20 degree earth elevation in a day

Maximum value of Doppler shift in term of eccentricity is shown in figure 9. By decreasing the eccentricity of this orbit, its circularity increase and satellite altitude in visibility durations goes down and hence Doppler frequency shift rises.

The maximum positive peak of this parameter is shown in figure 10.

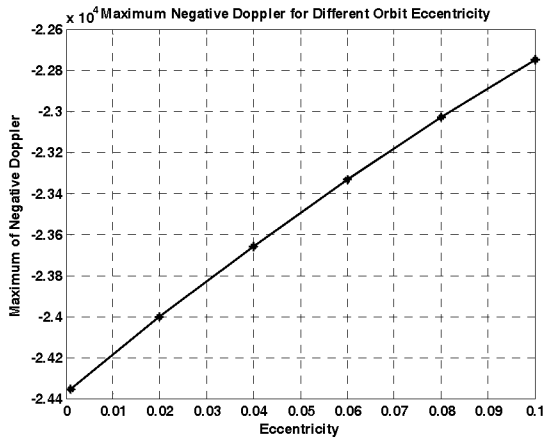


Fig. 9. Maximum negative Doppler in term of eccentricity

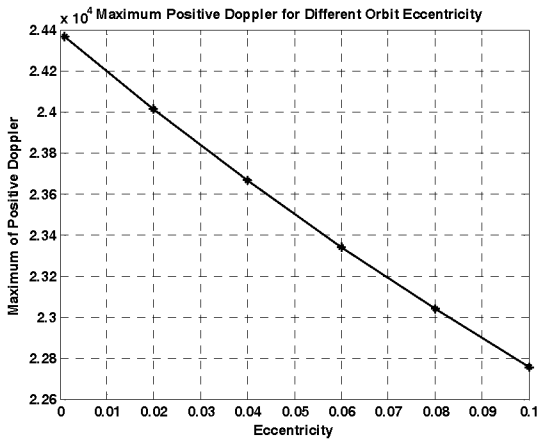


Fig. 10. Maximum positive Doppler frequency in term of eccentricity

Another important parameter is rate of Doppler shift, figure 11 shows the Doppler rate in the fourth pass. With a presumed carrier frequency of 1.2 GHz, Doppler rate is below 100 Hz/sec as seen in the figure.

Visibility interval is another parameter that is considered in the paper. The sum of visibility intervals in a day in term of various earth elevation angles is presented in figure 12.

VI. CONCLUSION

In this paper, we have proposed a Doppler prediction scheme with information of satellite orbital parameters and ground terminal position. This scheme can calculate Doppler shift for various kind of satellite orbits. Working with this estimator is very simple because all the orbital and ground station parameters can be put in MATLAB simulink environment. Another result obtained from this study is the range of Doppler frequency rate in the operation band. This parameter is needed to design the receiver phase lock loops, properly. For example, the DFR of a L-band transceiver with 1.2 GHz carrier frequency is about 100Hz.

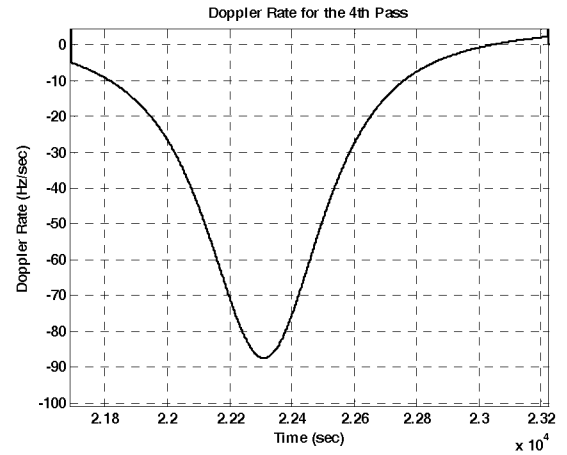


Fig. 11. Doppler rate in forth passes

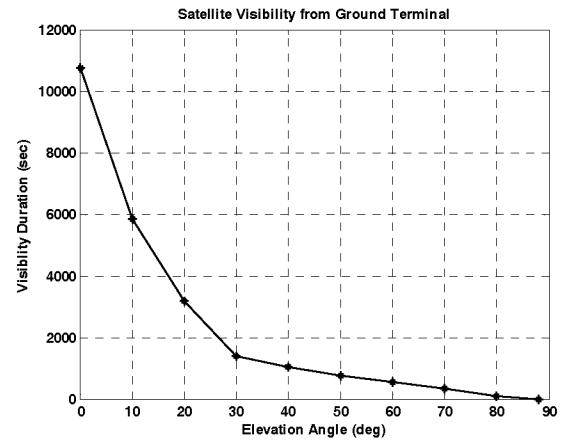


Fig. 12. Sum of visibility intervals in term of earth elevation angle

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